

Western Mathematics

TOPIC REVISION QUESTIONS

Mathematics Standard 2

MS-N2 Network Concepts

Section I – Multiple Choice Questions (1 Mark each)

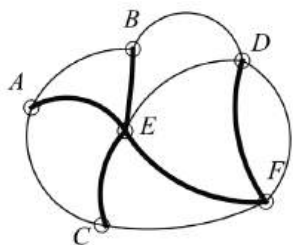
Section II – Longer Questions (Marks indicated beside questions)

Section I

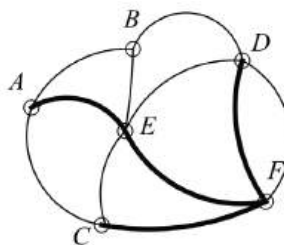
16 marks

1. Which diagram has a spanning tree for the network shown in bold?

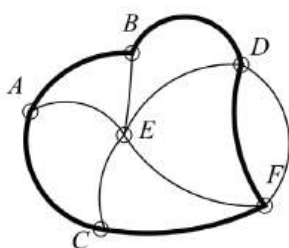
(A)



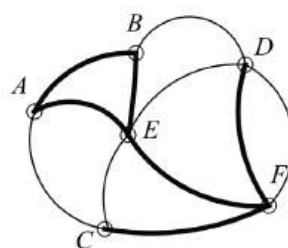
(B)



(C)

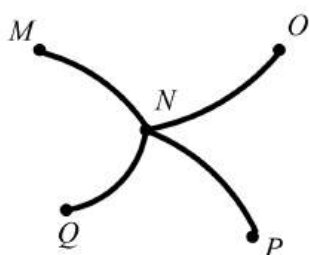


(D)

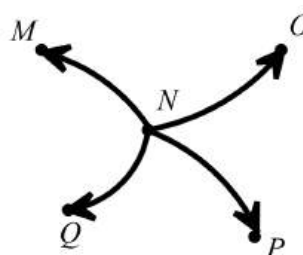


2. Which diagram shows a weighted, directed network?

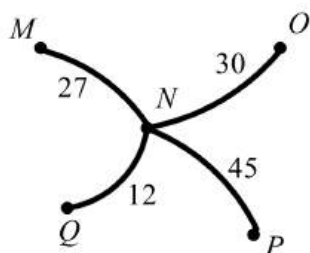
(A)



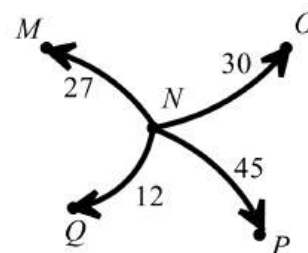
(B)



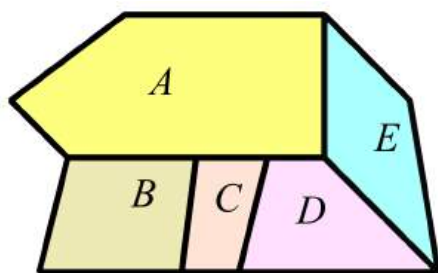
(C)



(D)



3. The diagram shows a map of five blocks of land which share some boundaries.

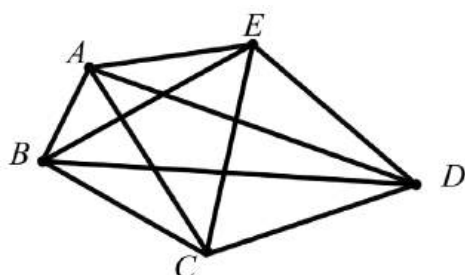


A network is drawn to represent the map.

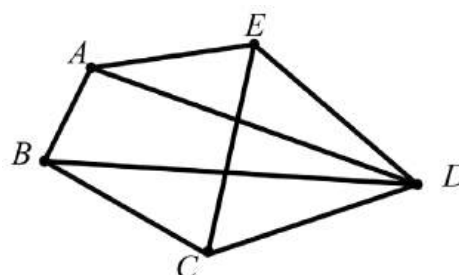
Vertices indicate blocks of land and edges indicate a shared boundary.

Which network represents the map?

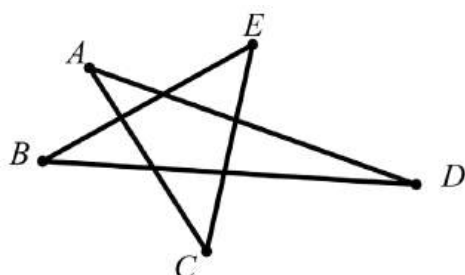
(A)



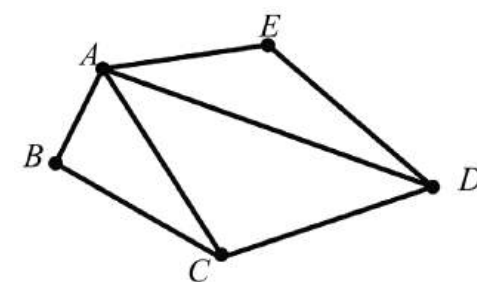
(B)



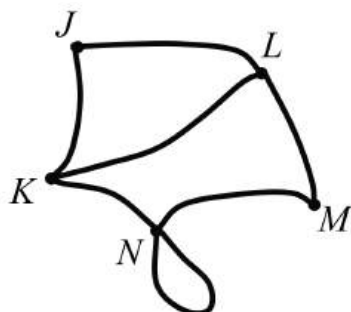
(C)



(D)



4. What is the degree of the vertex N in the network below?



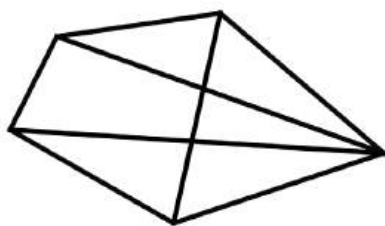
(A) 2

(B) 3

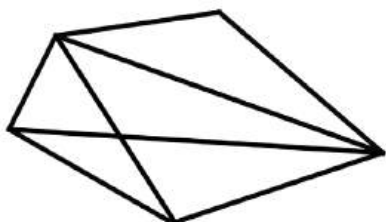
(C) 4

(D) 5

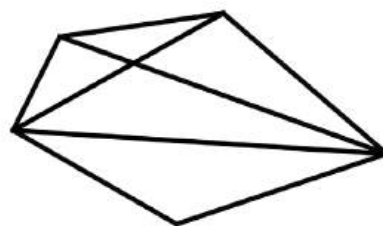
5. Which of the networks below is topologically equivalent to this network?



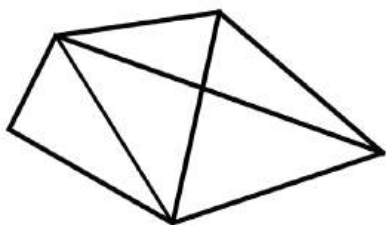
(A)



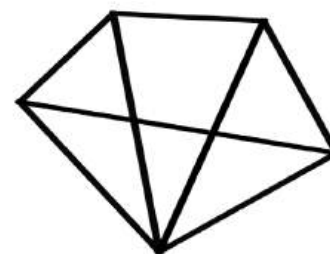
(B)



(C)



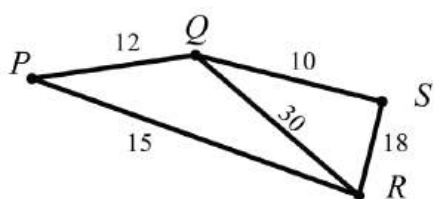
(D)



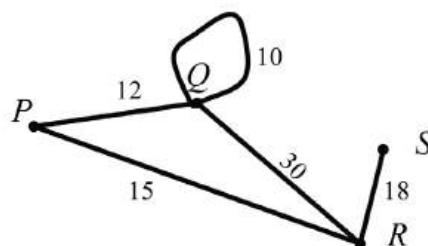
6. Which network is represented by the table below?

	P	Q	R	S
P	-	12	15	-
Q	12	10	30	-
R	15	30	-	18
S	-	-	18	-

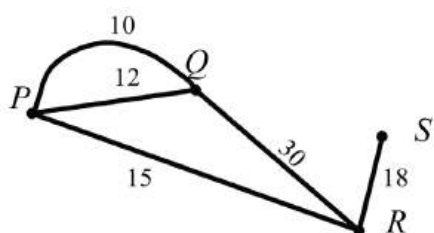
(A)



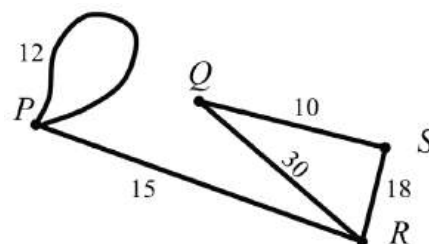
(B)



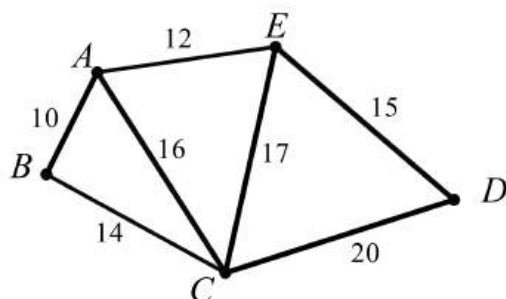
(C)



(D)



7. Which table could represent the network below?



(A)

	A	B	C	D	E
A	-	10	16	-	12
B	10	-	14	-	-
C	16	14	-	20	17
D	-	-	20	-	15
E	12	-	17	15	-

(B)

	A	B	C	D	E
A	-	10	16	-	12
B	10	5	14	-	-
C	16	14	-	20	17
D	-	-	20	5	15
E	12	-	17	15	-

(C)

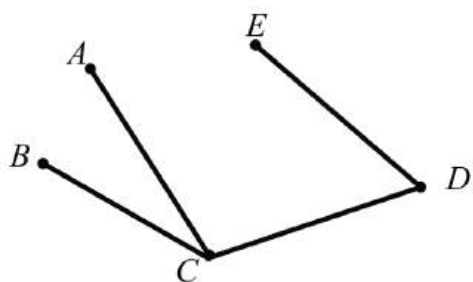
	A	B	C	D	E
A	-	15	17	-	12
B	15	-	20	-	-
C	17	20	-	14	16
D	-	-	14	-	10
E	12	-	16	10	-

(D)

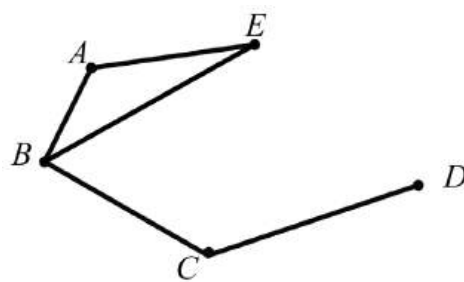
	A	B	C	D	E
A	-	10	16	-	12
B	-	-	14	-	-
C	-	-	-	20	17
D	-	-	-	-	15
E	-	-	-	-	-

8. Which diagram shows a network which forms a tree?

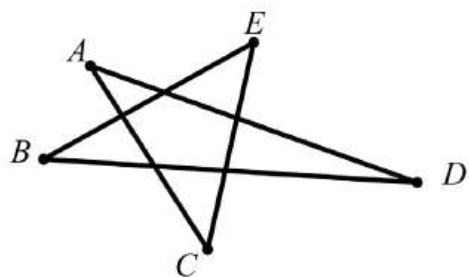
(A)



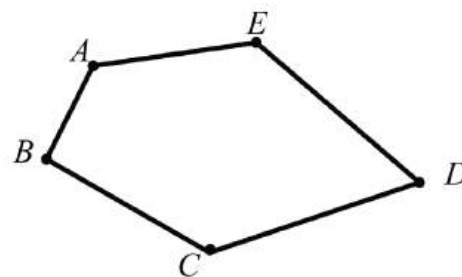
(B)



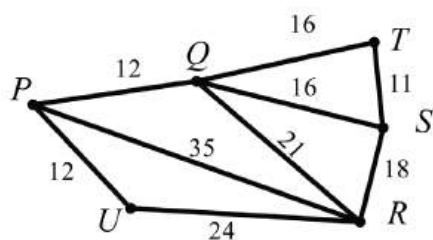
(C)



(D)



9. A minimum spanning tree is required for the network below.



Using Kruskal's algorithm, which would be the starting point?

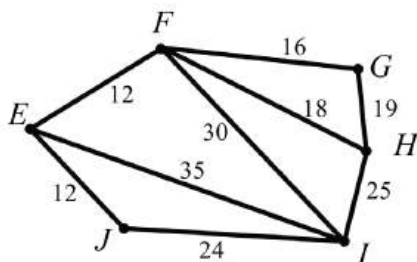
(A) The edge PU

(B) The edge TS

(C) The vertex P

(D) The vertex S

10. Kelly uses Prim's algorithm to find a minimum spanning tree for the network below. She chooses the vertex I as a starting point.

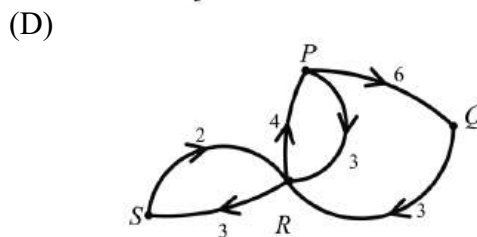
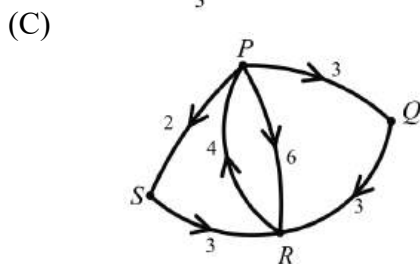
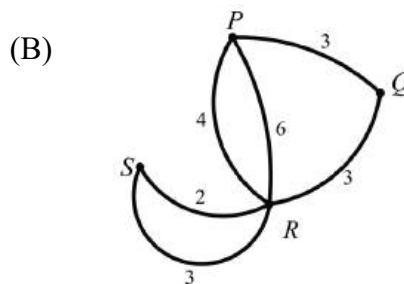
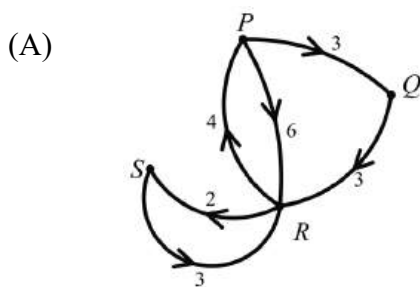


What would be the next vertex added to the spanning tree?

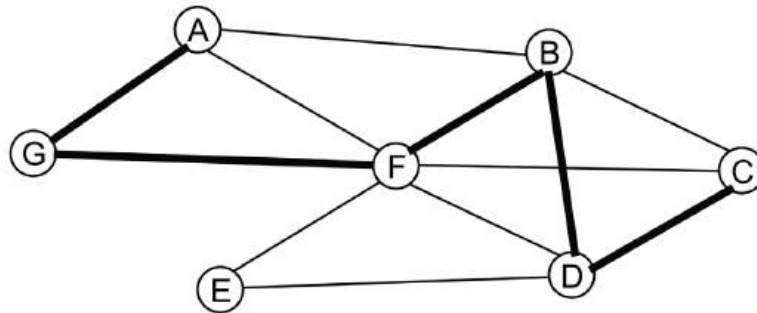
- (A) E
 (B) F
 (C) J
 (D) H
11. A directed network is to be drawn from the table below.

		To			
		P	Q	R	S
From	P	-	3	6	-
	Q	-	-	3	-
	R	4	-	-	2
	S	-	-	3	-

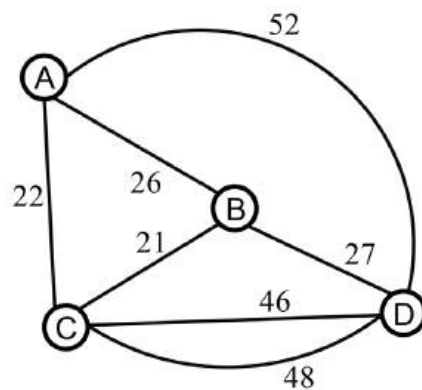
Which is the correct network?



12. What is shown in bold on this network diagram.

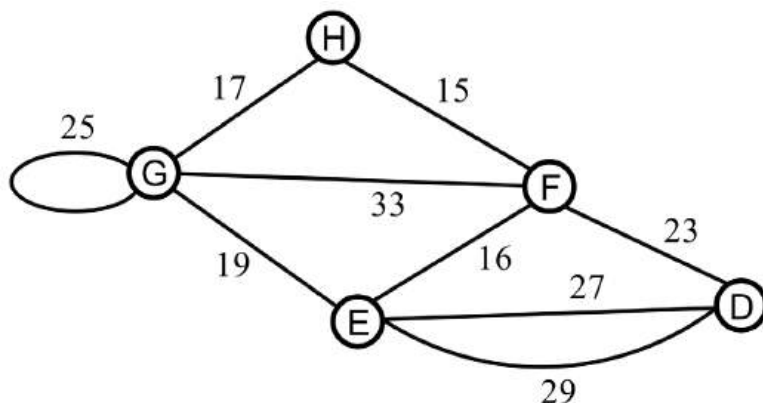


- (A) A cycle
(B) A disconnected network
(C) A path
(D) A spanning tree
13. What is the weight of the minimal spanning tree of this network?

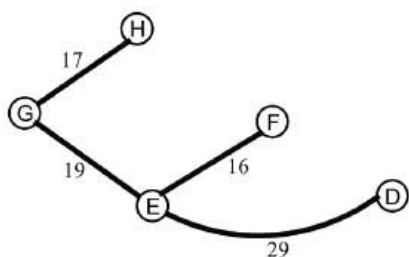


- (A) 53
(B) 69
(C) 70
(D) 75

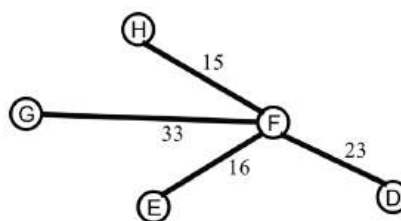
14. Which is a minimal spanning tree of this network?



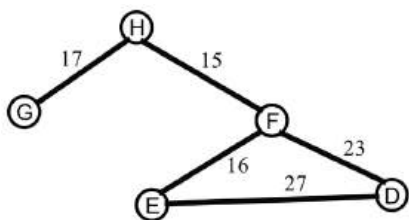
(A)



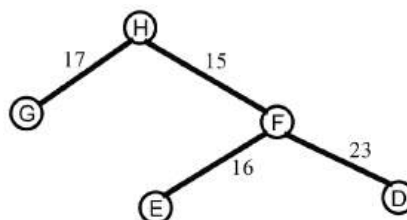
(B)



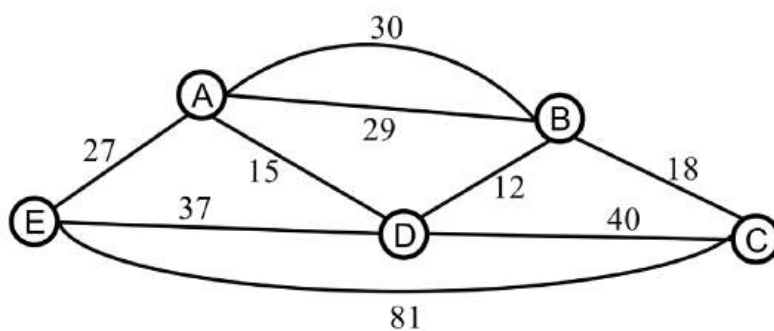
(C)



(D)



15. What is the shortest path from E to C on this network?



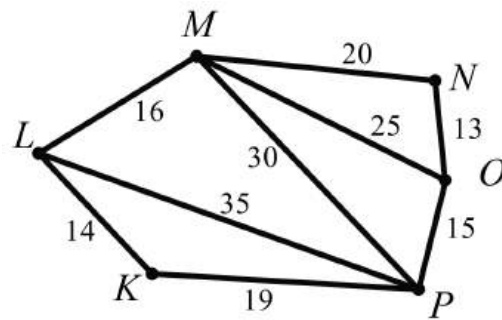
(A) EDC

(B) $EDBC$

(C) $EABC$

(D) $EADC$

16. Six towns, Kosmic, Lorde, Mire, Nico, Oscar and Price are connected by a network of roads as shown below. The travel times (in minutes) between the towns are given for each road.



What is the shortest travelling time from Lorde to Oscar?

- (A) 36 minutes
- (B) 41 minutes
- (C) 48 minutes
- (D) 49 minutes

Western Mathematics

Mathematics Standard 2

TOPIC REVISION – MS-N2 Network Concepts

QUESTIONS

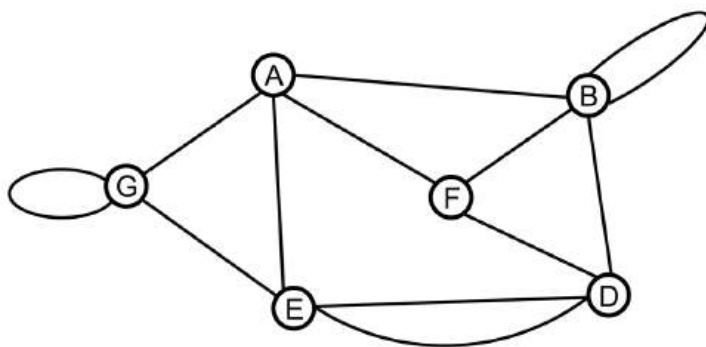
Section II

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 17 (4 marks)**Marks**

A network diagram is shown below.



- (a) How many loops are there in the network? 1

- (b) What is the degree of the vertex E ? 1

- (c) What term would describe a trip from F to B to D and back to F ? 1

- (d) Draw a spanning tree (by highlighting edges) on the network? 1

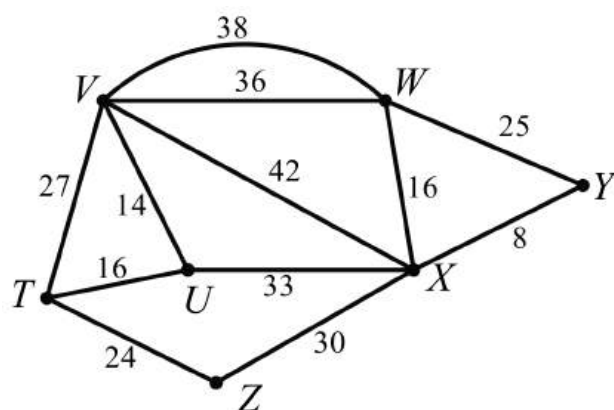
Marks

Question 18 (3 marks)

A new water supply network is to be constructed for seven suburbs.

The designers have proposed several possible routes for the pipes which are shown below with their costs *in thousands of dollars*.

Any of the suburbs can be the source for the network.



Determine the minimum cost of pipes needed to provide water to all suburbs.

3

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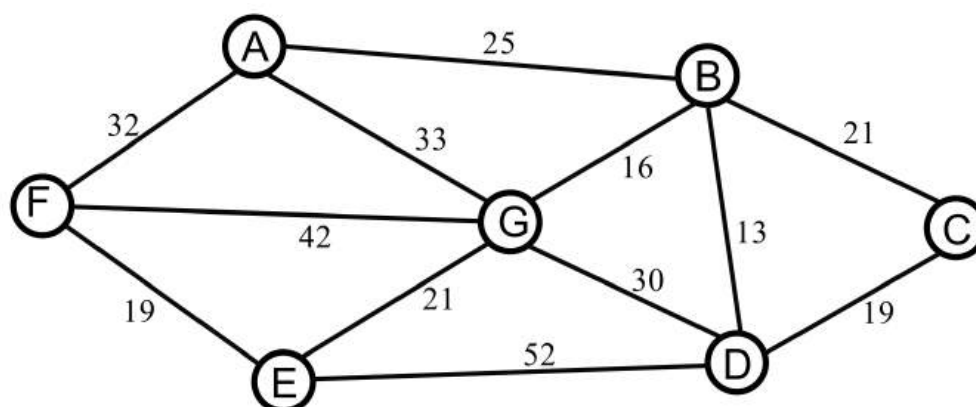
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Marks

Question 19 (4 marks)

The network diagram below shows the distances between seven villages in a valley.



(a) Complete the table to represent this network.

2

	A	B	C	D	E	F	G
A							
B							
C							
D							
E							
F							
G							

(b) Calculate the shortest distance between F and D.

2

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Marks**Question 20** (5 marks)

The table below represents a directed network of roads joining four towns Evans, Fuller, Green and Hall.

		To			
		E	F	G	H
From	E	-	50	18	-
	F	48	-	30	-
	G	-	33	-	16
	H	-	15	-	15

The weights represent the time of travel in minutes between each town in minutes.

(a) Draw a sketch of the network.

3

(b) Calculate the minimum travel time from Evans to Fuller.

2

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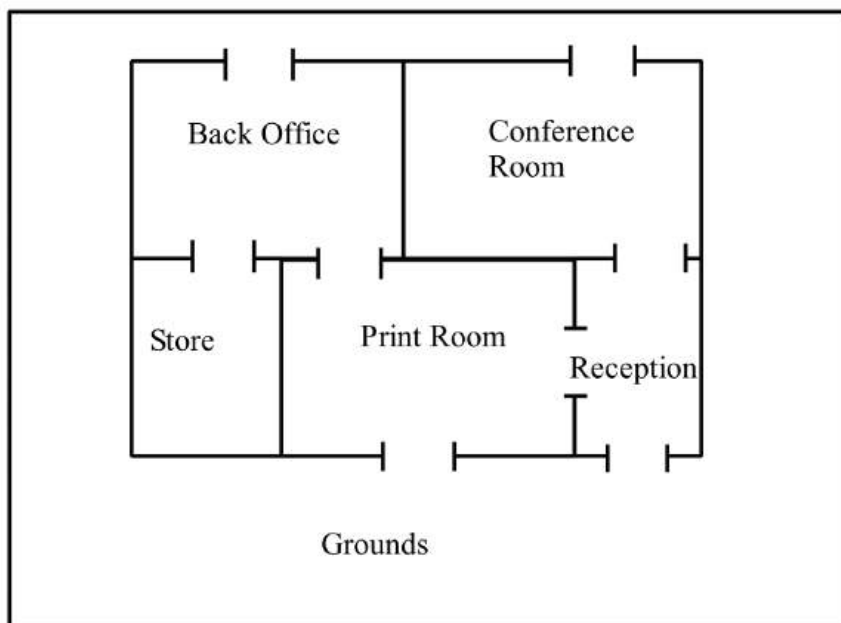
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Question 21 (6 marks)

The diagram below shows the plan of a business premises and the grounds on which they stand.



- (a) Using the rooms (and the grounds) as vertices, draw a network diagram from the plan. 2

Use the first letter as the label for each vertex. (e.g. *S* for the Store, *B* for Back Office)

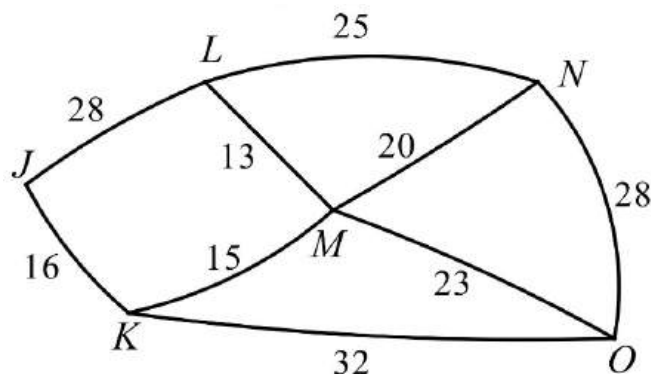
- (b) On your diagram highlight a spanning tree. 2

- (c) A path from Reception to the Store is *RPBS*. 2
List two other paths from Reception to the Store.

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Question 22 (4 marks)

A network of roads joining six towns is shown below.



- (a) How many vertices are of degree three (List them.)? 1

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- (b) Name a cycle which includes M and has four vertices. 1

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- (c) Mark a minimum spanning tree on the diagram and give its length. 2

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- (d) Find the shortest path from J to O and give its length. 2

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End of Revision

Western Mathematics

Mathematics

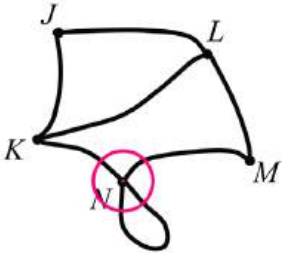
Standard 2

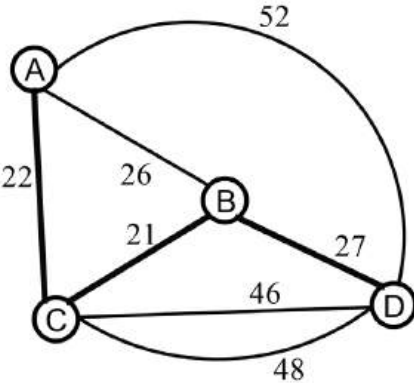
MS-N2 Network Concepts

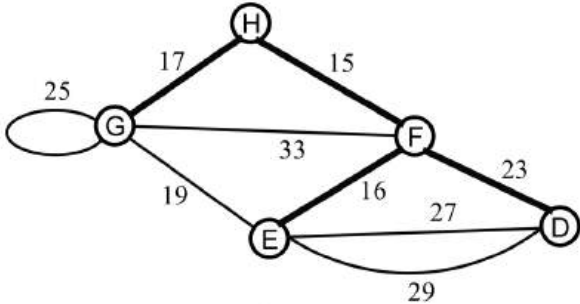
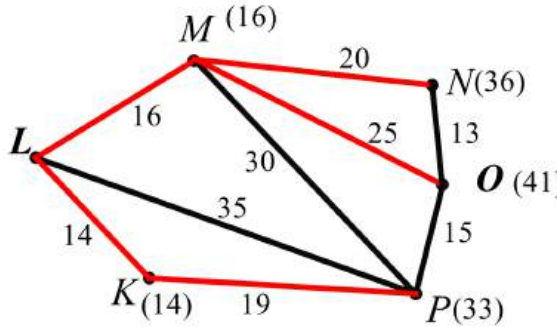
REVISION

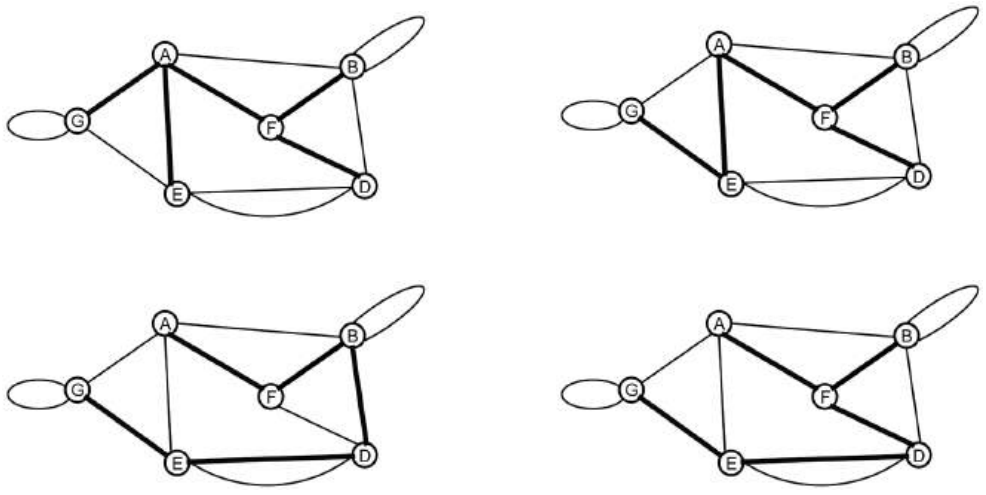
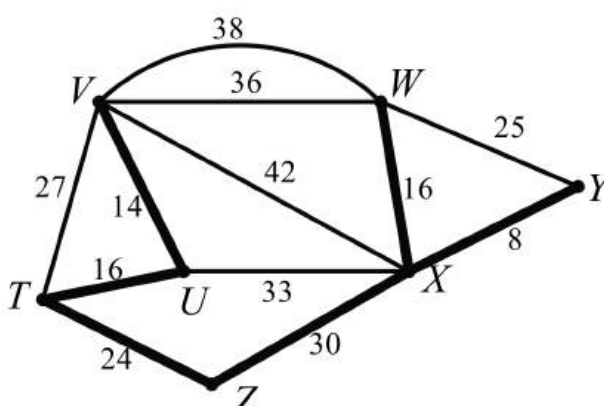
QUESTIONS

SOLUTIONS

Western Mathematics MS-N2 Network Concepts Revision Section I - Worked Solutions		
No	Working	Answer
1	<i>A spanning tree passes through every vertex and includes no loops or circuits. Only A does both.</i>	A
2	<i>A directed network has arrows to indicate direction and a weighted network has numbers on edges to indicate some value.</i> <i>Only D has both.</i>	D
3	<i>A has a gate to all other vertices, B, C D and E</i> <i>B also connects to C</i> <i>C also connects to D</i> <i>D also connects to E</i> <i>Only option D does all these</i>	D
4	 There are 4 lines that leave <i>N</i> so degree is 4.	C
5	The original network has 5 vertices and 8 edges The degree of the vertices are 3, 3, 3, 3, 4. All the other networks have 5 vertices and 8 edges. A. The degrees are 3, 3, 4, 2, 4. B. The degrees are 2, 4, 3, 3, 4. C. The degrees are 4, 2, 4, 3, 3. D. The degrees are 4, 3, 3, 3, 3. (Same as original in different positions)	D

6	From the table it is an undirected network, as the table is symmetrical. <i>P joins to Q with weight 12</i> <i>P joins to R with weight 15</i> <i>P joins to Q with weight 12</i> <i>There is a loop at Q with weight 10</i> <i>R joins to Q with weight 30</i> <i>S joins to R with weight 18</i> <i>Only option B meets all these</i>	B
7	<i>Option A is correct</i> <i>Option B has 5's on the diagonal, meaning there are two loops which don't exist</i> <i>Option C has the numbers in the wrong positions</i> <i>Option D has only one set of numbers, not the mirror image meaning it is a directed network.</i>	A
8	<i>B. Cand D all include cycles so are not trees, leaving A</i>	A
9	<i>uskal's algorithm starts with the lowest weight edge, so choose the edge TS.</i>	B
10	<i>Starting from I the lowest weight edge is IJ (24) so J would be the next vertex added.</i>	C
11	<i>B is not a directed network so eliminate it.</i> <i>D has the wrong combination of P to Q and P to R</i> <i>C does not have 2 connections form R to S</i> <i>A has all the correct connections in the correct directions</i>	A
12	<i>The highlighted section is not a cycle and is not disconnected. It is a path and a tree, but not a spanning tree as not all vertices are connected.</i>	C
13	 Using Kruskal's algorithm Choose BC first (21) then AC (22) reject AB as it creates a cycle choose BD (27) This then includes all vertices so is the minimal spanning tree Weight is $21 + 22 + 27 = 70$	C

14	 Using Prim's algorithm Choose F as start FH is first edge of tree Add FE as next lowest edge from F or H Add HG lowest connecting to tree Add FD as lowest connection to D	D
15	(A) $EDC = 37 + 40 = 77$ (B) $EDBC = 37 + 12 + 18 = 67$ (Shortest path) (C) $EABC = 27 + 29 + 18 = 74$ (D) $EADBC = 27 + 15 + 12 + 18 = 72$	B
16	 The shortest path is LMO which takes 41 minutes	B

	MS-N2 Network Concepts	Section II Solutions	Marks
Question 17			
(a)	2 (at <i>B</i> and at <i>G</i>)		1
(b)	4		1
(c)	A cycle (or a path)		1
(d)	Multiple possible answers (four are shown below)		1
			
Question 18			
 The minimum spanning tree is required. Using Kruskals algorithm, we start with XY (cost \$8K) Add VU (\$14K) Add TU (\$16K) Add WX (\$16K) Add TZ (\$24K) Reject VT and WY (creates cycles) Add ZX (\$30K) Which then connects all the vertices Cost = 14 + 16 + 24 + 30 + 16 + 8 = \$108K or \$108 000			3 marks

Question 19

(a)

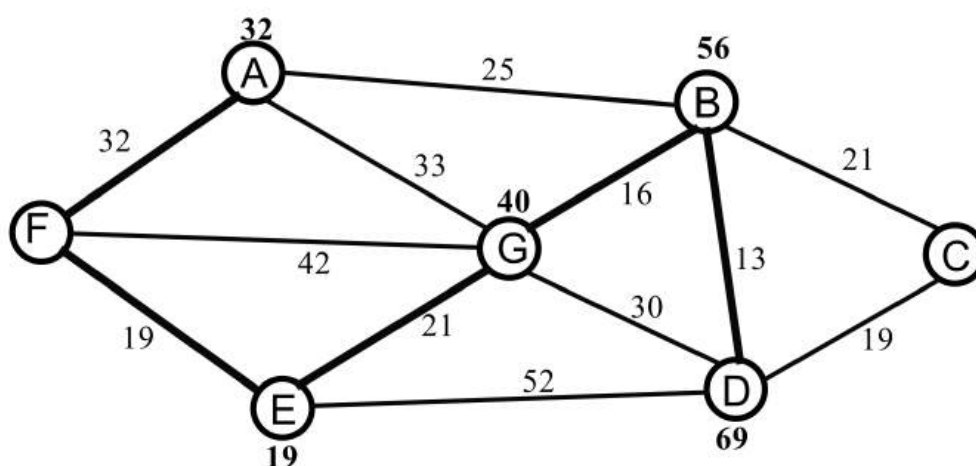
2 marks

	A	B	C	D	E	F	G
A	-	25		-	-	32	33
B	25	-	21	13	-	-	16
C	-	21	-	19	-	-	-
D	-	13	19	-	52	-	30
E	-	-	-	52	-	19	21
F	32	-	-	-	19	-	42
G	33	16	-	30	21	42	-

(b)

Applying Dijkstra's algorithm

2 marks



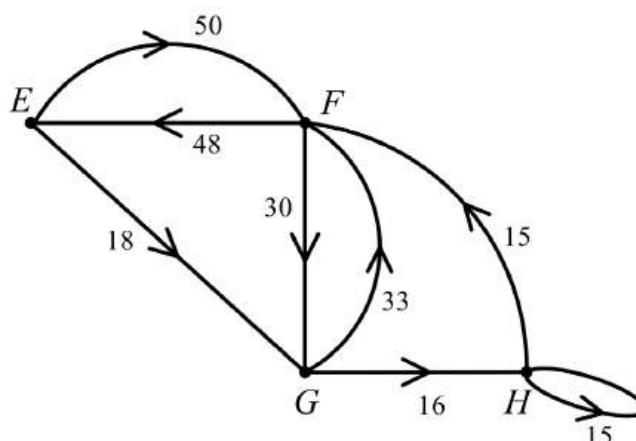
The shortest path is FEGBD at a distance of 69 km.

Question 20

(a)

3 marks

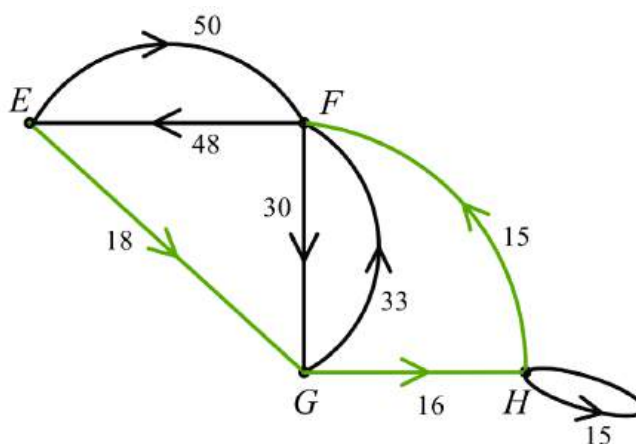
		To			
		E	F	G	H
From	E	-	5 0	1 8	-
	F	4 8	-	3 0	-
	G	-	3 3	-	1 6
	H	-	1 5	-	1 5



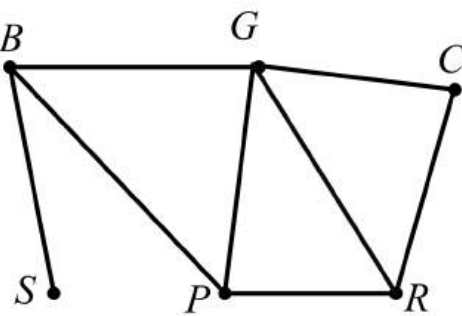
(b)

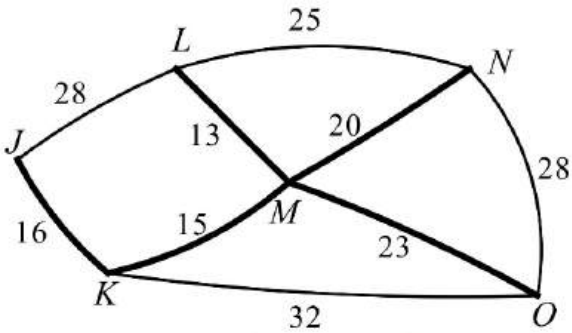
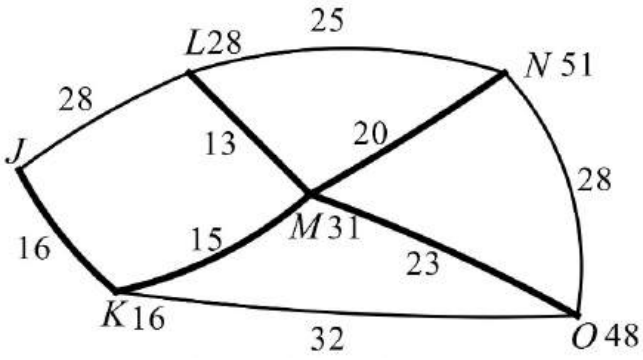
2 marks

E to F directly is 50 min

EGF is $18 + 33 = 51$ minEGHF is $18 + 16 + 15 = 49$ min

Question 21

(a)		2 marks
(b)	Two possible solutions are shown. 	2 marks
(c)	Any TWO of $RGBS$, $RPGBS$, $RGPBS$, $RCGBS$, $RCGPBS$.	2 marks

Question 22		
(a)	There are 4 (K, L, N and O)	1
(b)	$MLJKM$ or $MLNOM$ starting from any vertex and going clockwise or anticlockwise	1
(c)	Select LM – Least Select MK Select JK Select MN Select MO Stop as all vertices joined Length = $16+15+13+20+23 = 87$ 	2
(d)	Shortest path is JKO which is 48 minutes 	2